

CartoonFlix

Cartoon Movies App

CartoonFlix is a simple cartoon movies application developed using Flutter and Dart. The main idea of the project is to allow users to browse cartoon movies, search for movies, filter them by genre, and view more details about each movie.

The application also allows users to organize movies into Favorites, Want To Watch, Watching Now, and Watched lists.

Technologies Used

- **Flutter & Dart:** Used to build the application and its user interface.
- **TMDB API:** Used to get real movie data such as titles, posters, ratings, and descriptions.
- **Firestore Authentication:** Used for user signup, login, and logout.
- **Provider:** Used to manage the application state and update the UI when the movie data changes.
- **Hive:** Used as a local database to save the user's Favorites, Want To Watch, Watching Now, and Watched lists.

Project Structure

(organized into different folders to keep the code clear and easy to manage)

- **Screens:** Contains the application screens.
- **Services:** Contains the API code and Firebase Options.
- **Controllers:** Contains the main movie logic.
- **Providers:** Handles state management using Provider.
- **Models:** Contains the movie model.
- **Database:** Contains the Hive database code.

Main Features

Login and Signup

Users can create an account and log in using their email and password. Firebase Authentication is responsible for handling the authentication process.

Home Screen

The Home Screen displays the available cartoon movies in a grid and icons(Account, Favorites, Want To Watch, Watching Now, and Watched lists).

Search

The search bar allows users to search for a specific movie by its title.

Genre Filter

Users can filter the movies according to their genre, such as [Comedy, Adventure, Family, Fantasy, Action, and Music.](#)

Movie Details

When the user selects a movie, a details screen opens and displays [the movie poster, title, rating, and description.](#)

Favorites

Users can add movies to their Favorites list by pressing the heart icon. [They can also remove movies from the list.](#)

Watching Now , Watched and Want To Watch

Users can add movies to Watching Now or Want To Watch or mark them as Watched. [A movie cannot be in both lists at the same time.](#)

Data Storage

Hive was added to save the user's movie lists locally. This allows Favorites, Watching Now, and Watched movies to remain saved even after the user logs out or closes and opens the application again.

Conclusion

CartoonFlix helped me practice Flutter concepts in one project, including API calling, Firebase Authentication, Provider state management, and local data storage using Hive.